ACCRA-FLOW

A Modular Geospatial AI Framework for Paratransit Optimization and Urban Resilience in the Greater Accra Metropolitan Area

Case Study Group 5: Urbanization and Infrastructure Resilience
Adaboh Kwame Twum, Cajetan Songwae, Shaun Esua-Mensah, Pout Chop Madeng
Microsoft Mentors: Moses Igbinigie & Elan Jennings-Jones
Faculty Advisor: Dr. Nana Kwasi Karikari
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1. Executive Summary

Every day, Accra grinds to a halt. Not because the city lacks roads, but because it lacks information. Thousands of TroTros and Okadas, the minibuses and motorcycle taxis that carry the majority of Greater Accra's commuters, operate in near-total informational blindness. Drivers choose routes by instinct. Intersections treat a 28-passenger minibus and a private saloon car as equivalent. Commuters board with no idea when they will arrive. The annual cost: approximately $301.5 million, roughly 8.21% of Ghana's GDP. Accra-Flow asks a focused, urgent question: given the network that exists today, how much of that loss is recoverable through better information alone? The answer is substantial.

The framework gives drivers real-time demand signals, gives intersections the intelligence to prioritise a loaded minibus over an empty car, and gives commuters a reliable journey estimate. It does this through three layers: smart roadside cameras that process footage on-device to protect privacy; a self-learning signal system where intersections cooperate as autonomous agents to maximise passenger throughput; and a communication layer built for Accra's reality, USSD alerts that work on any handset, and mobile-money fuel rebates that reward driver cooperation directly.

2. The Problem Domain: An Economic and Human Emergency

2.1 The Scale of Congestion

Traffic congestion in Accra is not just an inconvenience: it is a structural drag on national development. The $301.5 million lost annually disaggregates into three channels:

Cost Category	Annual Impact (USD)	Share of Total
Time lost in traffic	$214.3 million	71%
Stress-driven productivity decline	$54.6 million	18%
Fuel waste from idling vehicles	$29.1 million	11%
Total annual loss	$301.5 million	100%

Table 1: Annual economic losses due to urban traffic congestion in Accra. Source: MyJoyOnline; cross-referenced with World Bank Urban Mobility Report.

Beyond economics, the crisis is a public health emergency. Air pollution from aged vehicle engines idling in traffic cost Ghana an estimated $1.6 billion in 2019. In the Kaneshie district, fine particulate matter (PM2.5) levels reach ten times the WHO 24-hour safe limit. Road-traffic injuries claim 10,000 lives annually and consume a further 1.6% of national GDP.

2.2 Who Bears the Burden

Aggregate figures obscure where the pain falls. Accra-Flow's design was anchored to four archetypes representing the modal experience of Greater Accra's commuter population:

Archetype	The Mechanism of Harm
Students (e.g., Abena, University of Ghana, Legon)	Loses approximately 10 hours per week to unpredictable journey times. Without real-time transit information, departure times are governed by

Archetype	The Mechanism of Harm
	guesswork, systematically eroding academic performance.
Perishable goods traders (e.g., Auntie Efua, Makola Market)	Reports up to 14.4% post-harvest loss due to transportation delays — a food security crisis as much as a mobility one.
TroTro drivers (e.g., Kojo)	Burns a significant share of daily gross revenue in idling fuel. Operating on thin margins in ageing vehicles, a single mechanical failure can tip a driver into debt.
Private commuters (e.g., Mensah, Airport Residential)	Loses an estimated 15% of productive commute time to congestion that coordinated signal priority could eliminate.

Table 2: Socio-economic archetypes illustrating the differential burden of congestion.

These archetypes became the design constraints that every technical decision was tested against: any solution that did not work on a basic feature phone, or that transferred financial risk onto a marginal driver, would fail at adoption.

3. Design Iterations

3.1 Iteration One: The Commuter Fluidity Index

The team's starting question was how to measure mobility quality in a system where a private car and a TroTro carrying 28 passengers share the same road. Conventional metrics treat all vehicles as functionally identical, analytically wrong in a paratransit context.

The response was the Commuter Fluidity Index (CFI): $CFI = \Sigma(P_i \times V_i) / (T_wait \times C_fuel)$. Each vehicle is weighted by its passenger count, giving a TroTro approximately fifteen times the mobility value of a private car.

Simulations exposed two limitations: the Balloon Effect (optimising a single intersection displaced bottlenecks rather than resolving them) and the Fixed-Route Problem (even where the CFI identified a faster detour, drivers would not take it without real-time demand visibility).

3.2 Iteration Two: The TroTro Priority Algorithm

The TroTro Priority Algorithm (TPA) replaced static traffic signals with smart signals that understand which vehicles matter most. A dynamic Priority Score is computed in real time combining occupancy-weighted vehicle counts, efficiency coefficients, starvation penalties, and touting penalties.

Despite its advances, the TPA remained a system of human-designed rules that could not adapt autonomously to conditions outside their design envelope, nor coordinate multiple intersections without a manually designed orchestration layer.

3.3 Iteration Three: Cooperative Multi-Agent Reinforcement Learning

In Accra-Flow, each intersection becomes an autonomous reinforcement learning agent. Under Multi-Agent RL (MARL), agents share information about their state and are rewarded for the health of the entire network:

$$\mathbf{R = w_1(CFI) + w_2(-D_queue) + w_3(-E_emissions) - \varphi(v_drift)}$$

The mechanism for solving the Balloon Effect is pressure propagation: each intersection periodically broadcasts its congestion pressure to its immediate neighbours. Coordination emerges from the learning process itself rather than from any static rule set.

4. Technology Stack

4.1 Tier 1: Perception

Roadside NVIDIA Jetson units are mounted at target intersections. The critical design choice is local processing: each unit analyses its camera feed on-device in real time, classifying vehicles by type, counting them, and tracking velocity without ever transmitting raw footage. Only anonymised metadata leaves the

device. Privacy compliance is built into the hardware layer as an engineering constraint, not a legal checkbox applied afterward.

4.2 Tier 2: Reasoning

This is where MARL agents live, where pressure propagation between intersections occurs, and where the system forecasts traffic conditions 5–30 minutes ahead. The geospatial foundation is OpenStreetMap, chosen because it is the only mapping platform that maintains stop-level data for informal TroTro networks.
A Synthetic-to-Real (S2R) pipeline trains MARL agents in high-fidelity simulations using SUMO (Simulation of Urban Mobility), DesRUTGe for realistic synthetic traffic generation, and NVIDIA Omniverse Digital Twins for risk-free testing before physical construction begins.

4.3 Tier 3: Interaction

USSD alerts require no internet connection, no data plan, and no smartphone. When the system identifies a demand hotspot or recommends an alternative route, the message reaches the driver's handset instantly and on any device.
The MoMo Fuel Rebate Token addresses a subtler barrier: for a driver whose fuel cost represents roughly 40% of gross revenue, an AI-recommended detour is a financial risk. The token system transfers that risk from the driver to the framework, converting driver behaviour from an obstacle into an asset.

5. Measuring Success: Pilot Metrics and Evaluation Framework

This section directly addresses how Accra-Flow will measure, validate, and demonstrate impact during and after a pilot deployment on the Madina-Tetteh Quarshie (N4) corridor.

5.1 Baseline Data Requirements

Before launching any pilot, the following baseline data must be collected over a minimum 4-week observation period:

- Average peak journey time on the Madina-Tetteh Quarshie corridor (measured via GPS-instrumented TroTros)
- Queue length and vehicle throughput at Oponglo Junction and Madina Police Station Junction (measured via roadside observation and initial Jetson unit deployment)
- Driver fuel expenditure as a percentage of daily gross revenue (measured via structured driver surveys and MoMo transaction histories)
- PM2.5 readings at 3 corridor monitoring points during peak hours
- TroTro passenger load factor (average occupancy per departure) at key hub stops
- Commuter journey time reliability: standard deviation of journey times across the pilot corridor

5.2 Pilot Metrics Dashboard

During the pilot phase, Accra-Flow will track the following primary and secondary metrics:

Metric	Baseline Target	Success Threshold (6 months)	Measurement Method
Average peak journey time, Madina-Tetteh Quarshie	67 minutes (SUMO-calibrated)	48-54 minutes (≥20% reduction)	GPS logs from 30+ instrumented TroTros; timestamped USSD check-ins
Vehicle throughput at pilot intersections	Baseline count TBD from Phase 1 observation	≥15% increase in peak-hour passenger-weighted throughput	NVIDIA Jetson occupancy tracking; CFI computed per 15-min interval
TroTro driver participation rate	0% (pre-launch)	≥60% of active N4 drivers enrolled in USSD/MoMo by Month 3	USSD registration logs; MoMo token redemption rate

Metric	Baseline Target	Success Threshold (6 months)	Measurement Method
Driver fuel cost as % of revenue	~40%	Reduction to 34-36% for participating drivers	MoMo transaction records; driver self-reported fuel receipts
USSD route recommendation adoption	Baseline N/A	≥40% of USSD-registered drivers follow ≥1 recommendation per week	System-side USSD interaction logs
Corridor PM2.5 index	Index 100 (baseline)	Index 78-85 (15-22% reduction)	Air quality sensors at 3 corridor points; correlated with idle-time logs
Journey time reliability	High variance (TBD)	20% reduction in standard deviation of journey times	GPS variance analysis across 30-day rolling windows
Commuter satisfaction	N/A (pre-pilot)	≥70% satisfaction among surveyed corridor commuters	Monthly commuter surveys at major stops (n≥150 per month)

Table 3: Pilot metrics, targets, and measurement methods.

5.3 Evaluation Timeframe

- Month 0-1: Baseline data collection; USSD gateway and MoMo token system load testing; driver recruitment
- Month 2-3: Phase 2 activation (Jetson units at Oponglo and Madina Police Station Junctions); first USSD recommendations issued; initial adoption metrics logged
- Month 4-6: Full metric evaluation against success thresholds; Ghost Bus simulation counterfactual audit to isolate causal effect of interventions
- Month 7+: Phase 3/4 expansion conditional on pilot thresholds being met at ≥4 of 8 primary metrics

5.4 What Success Looks Like at the End of a Pilot Phase

A successful pilot is not simply one that reduces journey times. It is one that demonstrates three things simultaneously:

- Technical validity: The MARL system demonstrably improves passenger-weighted throughput beyond what the baseline TPA rule-based system achieves alone
- Driver adoption: Participation rates above 60% validate the USSD/MoMo incentive design and confirm that the system is accessible to drivers without smartphones
- Scalability signal: System performance at two intersections does not degrade during high-variance events (rain, market days, junction reconfiguration), confirming readiness for Circle interchange expansion

6. Responsible AI: Beyond Privacy Compliance

Accra-Flow's Responsible AI framework was designed as a set of engineering constraints that shaped actual technical decisions, not a principles document consulted after the architecture was finalised. Each dimension below describes both the risk identified and the design choice made in response.

6.1 Privacy-by-Architecture

All video processing occurs on the NVIDIA Jetson edge device. No raw footage is ever transmitted to any server or stored beyond the processing cycle. The system's outputs are anonymised aggregates: vehicle counts, type classifications, velocity estimates, and occupancy scores. No individual journey history is created or retained. This design choice was made to comply with Ghana's Data Protection Act 2012 (Act 843) and to ensure community trust, a system that collected personal mobility data would face significant resistance from the communities whose cooperation the pilot requires.

6.2 Bias in Route Recommendations and Traffic Predictions

A geospatial AI system trained on existing traffic patterns risks systematically under-serving neighbourhoods that are already poorly connected. Accra-Flow addresses this through three mechanisms:

- Person-weighted optimisation: The Commuter Fluidity Index and MARL reward function optimise for passenger throughput, not vehicle count. This structurally prevents the system from over-prioritising private car corridors at the expense of high-occupancy TroTro routes.
- OpenStreetMap coverage auditing: Before deployment, the team will audit OSM stop-level data against ground-truth TroTro route maps to identify coverage gaps in peripheral neighbourhoods. Routes with incomplete mapping will be flagged and manually enriched before the pilot.
- Starvation penalty as anti-bias mechanism: The delta(t_wait) starvation penalty in the TPA/MARL reward function explicitly prevents side streets and lower-traffic corridors from perpetual deprioritisation. No intersection is allowed to wait indefinitely, regardless of the direction of AI optimisation.
- Post-deployment equity audit: After 3 months of pilot operation, the team will conduct a geographic equity audit comparing throughput improvements across corridor segments, checking whether gains are concentrated in already-favoured junctions or distributed across the N4 route.

6.3 Inclusion: Serving Users Without Smartphones or Data Plans

The digital divide in Greater Accra is not a peripheral edge case — it defines the majority of Accra-Flow's target users. The system's interaction tier was designed specifically for this:

- USSD as the primary driver interface: USSD works on any mobile handset, requires no data plan, and is familiar to Ghanaian users through airtime balance checks and mobile money. Route recommendations, demand alerts, and token confirmations are all USSD-native. No driver is required to own a smartphone to participate.
- No app requirement for commuters: Commuter journey time information will be disseminated through stop-based displays (Phase 3 target) and through partnerships with radio stations and community notice boards, channels that reach the non-smartphone population.
- MoMo as financial inclusion: Mobile money penetration in Ghana significantly exceeds smartphone penetration. Using MoMo wallets as the rebate mechanism means the fuel incentive reaches drivers who cannot use app-based systems.
- Offline resilience: Jetson units operate in standalone mode during power outages. Signal control defaults to a safe pre-programmed cycle rather than failing open. This is an inclusion consideration as much as a reliability one: low-income communities experience more frequent outages.

6.4 Safety Guardrails and Error Handling

Incorrect traffic guidance from an AI system is not a theoretical risk — it has direct safety implications in a dense urban environment with mixed vehicle types and pedestrian crossings. Accra-Flow's safety guardrails include:

- Conservative signal transitions: The MARL agent is constrained to a minimum all-red clearance interval between phase changes, preventing unsafe overlapping green phases regardless of optimization pressure.
- Confidence thresholds for USSD recommendations: Route recommendations are only issued when the system's 5–30-minute forecast confidence exceeds a defined threshold. Below that threshold, no recommendation is sent rather than sending a low-confidence one.
- Fallback to TPA rule-based control: If the MARL system's communication with the edge cloud is interrupted for more than 90 seconds, the Jetson unit reverts automatically to the TPA rule-based baseline. The system degrades gracefully rather than failing.
- Human override capability: Transport authority operators can issue manual override commands through the system dashboard at any time, maintaining human authority over the network.
- Touting penalty safety calibration: The drift touting penalty is applied only when a vehicle is detected crawling below 5 km/h outside a designated hub. The threshold was deliberately set to avoid penalizing legitimate slow movement in pedestrian-heavy zones.

6.5 Transparency and Explainability

Non-technical stakeholders — TroTro drivers, transport authority officials, community leaders — must be able to understand why the system makes the recommendations it does. Accra-Flow addresses this through:

- Plain-language USSD messages: Route recommendations include a single-sentence explanation ('High demand at Madina station, consider Adenta corridor, fuel credit GH₵2.00 applied on completion').
- Public performance dashboard: Monthly corridor performance data will be published and shared with the Greater Accra Metropolitan Assembly and relevant transport authorities, maintaining accountability to institutional partners.

7. Practical Constraints, Partnerships, and Stakeholder Buy-In

A technical sound system that cannot be deployed is not a solution. This section addresses the real-world constraints that determine whether Accra-Flow reaches the street.

7.1 Critical Institutional Partners

Partner	Role	Criticality
Ghana Highways Authority / GAMA	Road access rights for Jetson unit installation; signal infrastructure compatibility; corridor data sharing	Essential — without access rights, no hardware can be installed
Ghana Road Traffic Authority (DVLA)	Vehicle registration data for occupancy estimation calibration; regulatory approval for roadside hardware	High — required for legal deployment
MTN Ghana / Telecel Ghana	USSD gateway provisioning; MoMo API integration for fuel rebate tokens	Essential — USSD and MoMo are the primary driver interfaces
TroTro Driver Unions (GPRTU)	Driver recruitment for pilot; trust-building; co-design of USSD UX and rebate structure	Essential — without driver participation, the system has no last-mile effect
NVIDIA (Inception Program)	Jetson unit hardware access at subsidised research rates; technical support	High — hardware cost is the primary capital constraint
Ashesi University	Research validation; student field data collection team; ongoing algorithm refinement	Institutional anchor for academic credibility
Microsoft (AI for Good / Azure Credits)	Cloud compute for MARL training; Azure IoT Hub for edge-cloud messaging; programme credibility	High — compute costs are the primary operational constraint

Table 4: Critical partnership map for Accra-Flow pilot deployment.

7.2 Funding and Resource Requirements

The pilot phase (Phases 1-3, covering two N4 intersections over 6 months) has the following estimated resource requirements:

Resource	Estimated Cost / Source	Notes
NVIDIA Jetson Orin units (x4, including redundancy)	~$2,400 (hardware) + shipping; NVIDIA Inception Programme subsidy sought	2 units per intersection for redundancy
USSD gateway provisioning	~$500-800 setup; per-message costs covered by MoMo rebate programme budget	MTN Ghana partnership letter of intent to be secured
MoMo rebate token float (Phase 2 pilot)	~$3,000-5,000 for 3-month pilot assuming 60% driver participation at GH₵2/adherence event	Potential co-funding from transport operator associations

Resource	Estimated Cost / Source	Notes
Cloud compute (MARL training, Azure)	Microsoft AI for Good credit application submitted	Training compute is the largest ongoing operational cost
Field data collection (Phase 1 GPS instrumentation)	In-kind: Ashesi student team; 30 instrumented TroTros	Driver compensation: MoMo airtime credits
Air quality sensor network (x3 corridor points)	~$1,200-1,800; potential co-funding from Ghana EPA	Enhances Responsible AI environmental monitoring

Table 5: Pilot phase resource requirements and funding strategy.

Total estimated Phase 1-2 capital requirement: approximately $8,000-12,000 USD, with significant portions coverable through in-kind contributions, academic partnerships, and corporate social investment from MTN and NVIDIA. This is a deliberately conservative estimate that the team has scoped to be achievable without large institutional grants.

7.3 Securing Stakeholder Buy-In

TroTro Drivers

Drivers are the most critical and most sceptical stakeholder group. The team's approach to driver buy-in is based on four principles:

- Demonstrate before asking: The pilot will begin with a 2-week observation phase where drivers receive USSD information but are not yet asked to change routes. This builds familiarity with the system before financial incentives are introduced.
- Co-design the incentive: The MoMo rebate amount and redemption conditions will be co-designed with Ghana Private Road Transport Union (GPRTU) representatives before launch. Drivers will not be presented with a fait accompli.
- Risk transfer is explicit: Marketing materials for driver recruitment will lead with the fuel rebate guarantee — the system takes the financial risk of the recommended detour, not the driver.
- Peer champions: The team will recruit 5-10 early-adopter drivers as paid pilot champions to provide peer-to-peer advocacy within the driver community, leveraging existing trust networks.

Local Transport Authorities (GAMA, DVLA, Ghana Highways Authority)

- Frame as infrastructure investment: Accra-Flow is presented to authorities as an intelligent overlay on existing infrastructure, not a replacement of their systems or authority. The human override capability and public dashboard are specifically designed to maintain institutional control.
- Align with N4 expansion timeline: The Tetteh Quarshie interchange expansion is already underway. Accra-Flow positions itself as the information layer that maximizes the return on that infrastructure investment, a politically aligned message.
- Formal MoU before hardware installation: No Jetson units will be installed without a signed Memorandum of Understanding with the Ghana Highways Authority. This is a non-negotiable procedural step that builds institutional trust and legal clarity.

Non-Technical Judges and Decision-Makers

When presenting Accra-Flow to non-technical audiences, the team leads with the human impact before the technical architecture:

- The problem in one sentence: Accra loses $301.5 million every year because drivers, signals, and commuters have no shared information.
- The solution in one sentence: Accra-Flow gives those three groups the information they need, using technology already in drivers' pockets and on every street corner.
- The proof point: The Madina-Tetteh Quarshie corridor, where the team has already calibrated simulations against real traffic data, offers a concrete, visible, and politically significant test case.

7.4 Constraints That Could Block a Pilot and Navigation Strategies

Constraint	Risk Level	Navigation Strategy
Regulatory approval for roadside hardware installation	High	Engage Ghana Highways Authority through Ashesi University institutional channels in Month 1; frame as

Constraint	Risk Level	Navigation Strategy
		research pilot with defined removal date
Driver trust and participation below 60% threshold	High	GPRTU co-design; peer champions; 2-week observation-only phase before asking for route changes
MTN/Telecel USSD provisioning delays	Medium	Initiate partnership discussions immediately; identify Africa's Talking as a backup USSD aggregator (no direct carrier relationship required)
Power outages disrupting Jetson unit operation	Medium	Offline fallback to TPA rule-based control (already engineered); UPS battery backup on Jetson units
N4 corridor reconfiguration during interchange expansion	Medium	OSM geometries updated continuously; SUMO digital twin treats construction as a parameter, not a failure case
MARL training compute costs exceeding Azure credit allocation	Low-Medium	Training is front-loaded; Phase 2 onward uses pre-trained models with lightweight online learning

Table 6: Risk register and navigation strategies.

8. Projected Impact

Accra-Flow's projected outcomes are derived from the SUMO simulation environment calibrated against field measurements on the N4 corridor and from published evidence on comparable interventions in peer cities.

Metric	Baseline	Projected Post-Deployment	Basis
Average peak journey time, Madina-Tetteh Quarshie	67 minutes	48-54 minutes	SUMO simulation, N4 calibration
Annual economic loss recoverable	$301.5 million total	$45-72 million in Phase 1-2 scope	15-24% throughput improvement applied to time and fuel cost categories
PM2.5 contribution from idling (corridor)	Index 100	Index 78-85	Reduced idle time modelled against emissions coefficients
TroTro driver fuel expenditure as share of revenue	~40%	~34-36%	MoMo rebate coverage plus reduced idling time

Table 7: Projected impact metrics. Projections assume 60% driver participation and do not account for network effects from full corridor coordination.

9. Implementation Roadmap: Madina-Tetteh Quarshie Corridor

Phase	Activity	Key Metrics Activated
Phase 1: Grounding (Months 1-2)	Real-time GPS data from instrumented TroTros. SUMO digital twin calibrated. OSM geometries validated. Baseline data collection for all 8 pilot metrics.	Baseline journey time, throughput, PM2.5, driver fuel cost
Phase 2: Logic Deployment (Months 2-4)	NVIDIA Jetson units deployed at Oponglo and Madina Police Station junctions. TPA activated as rule-based baseline. USSD gateway and MoMo token validated end-to-end. Driver recruitment and onboarding.	USSD adoption, driver participation rate, throughput at intersections
Phase 3: Stress Testing (Months 4-6)	Ghost Buses injected into SUMO simulation as counterfactual audit tools. Full metric evaluation	Full metric dashboard; equity analysis; MARL vs

Phase	Activity	Key Metrics Activated
	against success thresholds. Geographic equity audit.	TPA comparative
Phase 4: Expansion (Month 7+)	37 Military Hospital Junction (emergency-vehicle priority). Circle (Kwame Nkrumah Interchange) as scalability validation.	City-scale throughput; Phase 4 unlocked only if Phase 3 meets ≥4/8 success thresholds

Table 8: Four-phase implementation roadmap.

10. Comparative Evidence and Policy Framework

10.1 Lessons from Peer Cities

The Digital Matatus project in Nairobi mapped the routes and stops of that city's informal minibus network and released the data openly. Local entrepreneurs built routing applications without coordination, creating a self-sustaining data ecosystem. Accra-Flow adopts the same open-data philosophy.

Research on Kigali's bus network found that performance-based driver compensation leads to higher passenger turnover but at the cost of faster, less safe driving. This finding directly shaped the MoMo incentive design: tokens reward adherence to route and safety-consistent speed profiles, not passenger counts.

10.2 Data Governance and Sovereignty

The Accra-Flow architecture was designed from the outset for compliance with Ghana's Data Protection Act 2012 (Act 843). All video processing occurs on the edge device; no raw footage is ever transmitted. Every component uses open-source or open-data alternatives, ensuring that no single vendor's pricing decision or service discontinuation can compromise the system. Greater Accra's urban mobility data is treated as infrastructure data whose governance must remain under Ghanaian institutional oversight.

11. Conclusion

The Accra-Flow development journey followed a consistent pattern: each iteration revealed not a flaw in the solution, but an underestimation of the problem's scope. The CFI correctly identified people as the right unit of optimisation. The TPA translated that insight into actionable signal logic. The MARL system transcends these constraints by allowing the network to discover optimal coordination policies through experience.

The framework's central claim is validated by the architecture itself: Accra's transit crisis is an informational failure masquerading as an infrastructure problem. The infrastructure was always there. What was missing was the intelligence to coordinate it.

The $301.5 million to be reclaimed annually is not merely a financial figure. It represents ten hours per week returned to students like Abena, perishable inventory preserved for traders like Auntie Efua, and fuel margins restored to drivers like Kojo. When urban mobility infrastructure fails, it fails people. When it succeeds, it returns to them the most finite resource they have: time.

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